

ANDREY OSIPOV

Python Backend Engineer

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GENERAL INFORMATION

Employment: Full-time

Work format: Remote, hybrid, on-site

Location: Belgrade, Serbia

Citizenship: Russia, Serbian residence permit

SUMMARY

Python backend engineer with 3+ years of commercial experience. Specialized in high-load services on FastAPI and Django: performance optimization, large-scale data, testing, observability, CI/CD and deployment. Own services end to end – from database schema and API design to infrastructure and production. Degree in information security – I write code with security and load in mind.

EXPERIENCE

Total commercial experience – 3+ years.

Safe Transport Innovation Center

Sep 2025 – Present

Python Developer

Remote

Services for Moscow transport-security systems. Sensitive data under NDA.

- Designed and shipped an aggregated data model over a multi-terabyte MongoDB store (**8.6B records**), consolidated data from dozens of collections, cut time to retrieve a full object summary **from 3.2s to 87ms** and reduced database load **by 37%**.
- Built from scratch a **Git-style state-versioning system** for the core service (change commits, state diffs, stable versions and a history graph), made regressions between releases traceable and sped up acceptance testing **1.7x**.
- Stabilized and extended a **high-load parallel data-processing system** (threading, multiprocessing) and ensured correct service behavior **under peak load**.
- Introduced **centralized error monitoring (Sentry)** and uptime alerting to Telegram across my services, cut mean time to detect incidents **from 4 hours to 5 minutes** and eliminated silent failures.
- Rewrote legacy services on FastAPI with an async architecture, reduced p95 latency **from 480ms to 85ms** and increased throughput **from 450 to 720 RPS**.
- Applied information-security practices in services handling sensitive data under NDA: input validation, access control and dependency review, which let the services **pass external security audits**.

Stack: Python, FastAPI, MongoDB, Redis, ClickHouse, Docker, RabbitMQ, asyncio, threading, multiprocessing, Sentry, Claude Code

Rutube

Sep 2024 – Aug 2025

Python Developer

Moscow

Leading Russian video platform. Django (DRF) monolith plus aiohttp and FastAPI microservices; team of 13 engineers; owned my services end to end – from development and code review to running them in production.

- Optimized **10+ high-load API endpoints** using Prometheus and Grafana data: reworked ORM queries (select_related/prefetch, indexes), added Redis caching and rate limiting and cut p95 latency **from 1.1s to 92ms** at **~16,000 RPS**.
- Rewrote heavy SQL queries using Django CTEs and window functions, cut their execution time **from 12 minutes to 8 seconds** and removed critical load from PostgreSQL.
- Moved **200+ tests** to pytest with parallelized CI/CD runs via pytest-xdist, raised coverage **from 42% to 78%** and significantly sped up the pipeline.
- Migrated API docs to OpenAPI 3.0, replaced django-rest-swagger with drf-spectacular, documented **157 endpoints** and delivered clear API contracts to neighboring teams.

- Ran a security audit of the services, closed **OWASP Top 10 vulnerabilities**, updated insecure dependencies and tightened access policies.
- Unified Dockerfiles for two services, adopted the uv package manager and cut CI image build time **from 8.5 minutes to 2 minutes** on average.

Stack: Python, Django, DRF, aiohttp, FastAPI, PostgreSQL, Redis, Prometheus, Grafana, Elasticsearch, Docker, pytest, microservices

PSK SOFT LLC

Python Developer

May 2023 – Aug 2024

Moscow

Backend development of corporate web services on Python/Django with PostgreSQL: server-side logic, integrations and production infrastructure.

- Built and shipped the **backend of 4 corporate web projects** on Python/Django (DRF) with PostgreSQL, implementing domain models, business logic and admin panels.
- Implemented server-side handling of client requests with validation, PostgreSQL storage and instant notifications and cut business response time **from 2 hours to 90 seconds**.
- Integrated **payment systems, web analytics and email delivery**.
- Ran initial performance optimization: added caching, reworked indexes and SQL queries and sped up page response **from 1.8s to 400ms**.

Stack: Python, Django, DRF, PostgreSQL, Nginx, Docker

TECHNICAL SKILLS

Languages: Python, SQL, Bash

Backend: FastAPI, Django, Django REST Framework, aiohttp, SQLAlchemy, Alembic, Pydantic, Celery, REST API, gRPC, OpenAPI/Swagger, JWT/OAuth, asyncio, threading, multiprocessing, pytest

Databases: PostgreSQL, MongoDB, Redis, ClickHouse, Elasticsearch, Apache Cassandra

Messaging & Brokers: Apache Kafka, RabbitMQ

DevOps & Infrastructure: Docker, Docker Compose, Kubernetes, Nginx, CI/CD (GitHub Actions, GitLab CI), Linux, S3

Monitoring: Prometheus, Grafana, Sentry

Other: Git, Agile, microservices

EDUCATION

BSc, Information Security – National Research Nuclear University MEPhI, Moscow

2025

Faculty of Cybernetics and Information Security

LANGUAGES

Russian – Native English – B2+ (Upper-Intermediate), certified (Cambridge FCE)